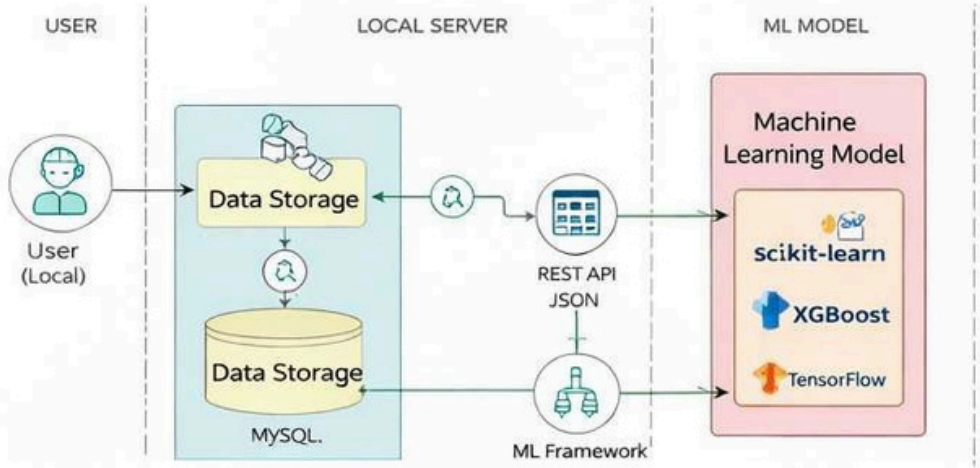
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	8 March 2026
Team ID	Team 199
Project Name	Insurance Fraud Detection Using Machine Learning
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table 1 & table 2

Example: Insurance Fraud Detection Using Machine Learning Stack



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

S.No	Component	Description	Technology
1	User Interface	How user interacts with application e.g. Web UI	Flask, HTML, CSS, JavaScript
2	Data Storage	Where collected claim data and processed features are stored	MySQL, PostgreSQL
3	Machine Learning Model	Fraud detection logic and algorithms	Scikit-learn, XGBoost, TensorFlow
4	API Integration	External and internal APIs for data exchange	REST API, JSON
5	ML Framework	Supports deployment and inference of ML.m models	Flask, Joblib, ONNX

Table-1: Technology Stack

S.No	Component	Description	Technology
2	Application Logic-1	Backend logic for processing insurance claim data and handling system requests	Python, Flask
3	Application Logic-2	Data preprocessing and feature engineering for insurance claim dataset	Python, Pandas, NumPy
4	Application Logic-3	Fraud detection prediction logic using trained machine learning model	Scikit-learn, XGBoost
5	Database	Stores insurance claim data, user information and prediction results	MySQL / PostgreSQL
6	Cloud Database	Optional cloud storage for storing large datasets and model outputs	AWS RDS / MongoDB Atlas
7	File Storage	Stores datasets, trained models and generated reports	Local Storage / AWS S3
8	External API-1	API used for fetching additional claim verification data	REST API
9	External API-2	API used for system integration or authentication	Google Auth API
10	Machine Learning Model	Detects fraudulent insurance claims based on historical data	Logistic Regression, Random Forest, XGGoost
11.	Infrastructure (Server / Cloud)	Deployment environment where the ML model and application run	Local Server / AWS / Docker

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used to develop backend and ML model	Flask, Scikit-learn, Pandas
2	Security Implementations	Secure authentication, data encryption and access control	HTTPS, JWT Authentication
3	Scalable Architecture	System designed to handle increasing claim data and uses	Microservice / Cloud Deployment
4	Availability	System available for fraud analysis whenever required	Cloud Hosting / AWS
5	Performance	Fast processing of insurance claim data and fraud prediction	Optimized ML models, Caching

References

- https://www.sciencedirect.com/science/article/abs/pii/S0275531922001325
- https://www.mdpi.com/2227-9091/11/9/160
- https://www.ijraset.com/research-paper/insurance-fraud-detection-using-machine-learning
- https://www.ijcaonline.org/archives/volume187/number65/machine-learning-driven-detection-of-fraudulent-vehicle-insurance-claims/
- https://www.researchgate.net/publication/367795142_Prediction_of_Insurance_Fraud_Detection_using_Machine_Learning_Algorithms
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